ARRIVAL Protocol

Structured Semantic Coordination Enables Sustainable Cross-Architecture AI Cooperation

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1. Abstract

We present the Arrival Protocol (Atomic Reasoning via Rival Validation and Adversarial Logic), a structured communication framework enabling AI-to-AI coordination across heterogeneous large language model (LLM) architectures without fine-tuning, shared weights, or prior joint training. The protocol employs 66 semantic @-atoms from DEUS.PROTOCOL v0.5, injected via system prompts, to establish a shared coordination vocabulary.

Across 2,200+ experiments organized into 23 phases involving 17 distinct LLM architectures, the principal findings are:

- **GPQA Diamond (N=40, Phase 16):** 65.0% accuracy with a homogeneous 5-agent Qwen3-235B ensemble vs. 52.5% majority voting (+12.5 pp, McNemar $p = 0.006$).
- **Full GPQA Diamond (N=198, Phase 20):** Arrival MV 67.7% vs. Solo CoT 62.1% (+5.6 pp, $p = 0.233$, ns).
- **Frontier models (Phase 21):** Non-Debate MV 82.3% vs. Arrival MV 78.8% (−3.5 pp). Anchoring identified as primary degradation mechanism.
- **CGD (Phase 22, N=198): 86.4%** accuracy (171/198, 95% CI [80.9%, 90.5%]); +7.6 pp over Arrival MV ($p = 0.009$); 61.6% questions unanimous (95.9% acc.).
- **CGD-7 (Phase 23, N=198):** 86.9% with 7 cross-vendor models; best solo Grok 4.1 87.4% outperforms ensemble — quality > quantity.
- **Ablation (WMV+D-3):** Pruned 4-model subset achieves **88.4%** (175/198) — the project-high accuracy.
- **Applied tasks (Phase 18):** 5-model swarm matches Claude Sonnet 4.5 on security audit (10/10 bugs) at comparable cost.
- **GovSim (Phase 19):** 0% → 100% survival (Fisher $p = 0.008$); CARE = 0.97, Gini = 0.001. Total cost: under $95 USD across ∼14,500 API calls.

2. Introduction

2.1 Problem Statement

The proliferation of large language models from competing vendors has created a fragmented ecosystem in which models cannot natively coordinate. Each architecture processes language through different tokenization, attention mechanisms, training corpora, and alignment. Yet many applications demand coordination: multi-agent reasoning, ensemble decision-making, negotiation, and collaborative problem-solving.

Recent work on multi-agent debate (MAD) has demonstrated that structured interaction between LLMs can improve factuality and reasoning [8]. However, the MAD paradigm faces critical challenges: unstructured debate can *degrade* reasoning quality [28]; strong single-agent prompting can match multi-agent systems on many benchmarks [31]; and existing MAS-LLM systems generally lack formal foundations, statistical rigor, and honest engagement with failure modes [19].

The Arrival Protocol addresses these challenges through three complementary innovations:

1. **Structured semantic atoms**: 66 compositional @-atoms provide a formal coordination vocabulary, preventing quality degradation.
2. **CRDT-based conflict resolution**: MEANING-CRDT v1.1 formalizes consensus as weighted vector averaging with proven convergence, bounded debt, and Bayesian equivalence.
3. **Empirical rigor**: 2,200+ experiments across 17 architectures with significance testing, adversarial testing, ablation, and engagement with critical work.

2.2 Contributions

- **Arrival Protocol**: 4-round structured coordination using 66 semantic atoms, validated across 17 LLM architectures.
- **MEANING-CRDT v1.1**: Mathematical framework with 11 theorems for semantic conflict resolution.
- **Echo-chamber analysis**: 7 quantitative metrics for detecting social conformity in LLM ensembles.
- **Hypercorrection paradox**: Naive memory degrades multi-agent accuracy; gated CARE-ALERT restores it.
- **Applied validation**: 5-model swarm matches frontier solo at comparable cost.
- **CARE as task discriminator**: Quantitatively distinguishes analytical (CARE ≈ 0.50) from constructive (CARE ≈ 0.97) tasks.
- **Cooperation breakthrough**: First structured multi-agent protocol solving tragedy of the commons for LLMs (0%→100% survival in GovSim).
- **CGD**: Agreement-gated debate achieving 86.4% on GPQA Diamond with mid-tier models.
- **Non-monotonic scaling**: Quality > quantity — pruned 4-model subset outperforms 7-model ensemble.
- **Comprehensive evaluation**: 2,200+ experiments, under $95, with 34 references to related work.

3. Related Work

3.1 Multi-Agent Debate

Du et al. [8] established that multi-agent debate improves factuality, founding the MAD paradigm. Liang et al. [22] introduced structured critique rounds. Li et al. [21] showed sparse communication topologies outperform fully connected ones. RUMAD [30] unifies debate with reinforcement learning. A-HMAD [34] proposes adaptive heterogeneous debate. ReConcile [5] demonstrates round-table consensus among diverse LLMs.

MoA [29] demonstrated that layered aggregation outperforms individual models. Li et al. [20] showed homogeneous ensembles can outperform heterogeneous ones — confirmed by our Phase 16. Hegazy [12] examines when smaller models debate larger ones. Ashery et al. [3] study social convention formation in LLM populations.

3.2 Critical Perspectives

Wang et al. [31] showed single-agent CoT can match multi-agent systems. Smit et al. [28] showed debate can *degrade* reasoning. Wu et al. [32] question whether LLM agents can really debate in a controlled logical reasoning study. La Malfa et al. [19] critiqued MAS-LLM research for lacking formal foundations. Choi et al. [6] proved a **martingale property**: debate does not change the expected correctness of LLM responses — validating our Phase 21 finding that debate is net zero on MCQ accuracy.

3.3 Game-Theoretic Cooperation

GovSim [23]: 12/15 LLM models achieve 0% survival in commons games; reproduced by Curvo et al. [7]. FAIRGAME [4]: end-game defection. Piché et al. [24]: RL shifts toward defection. Kaesberg et al. [14] investigate LLMs' failure to avoid tragedy of commons. Akata et al. [2] study repeated games with LLMs. SanctSim [26] simulates effects of sanctions. Hintze & Adami [13]:

cooperation emerges at a tipping point. Kostka & Chudziak [18] evaluate theory of mind in multi-agent systems.

3.4 Selective Debate and Confidence Gating

Fan et al. [10] propose **iMAD**: ML-trained confidence classifiers to gate debate. Eo et al. [9] examine whether natural language is optimal for debate (**DOWN**). Kaesberg et al. [15] systematically compare voting versus consensus. Ai et al. [1] go beyond majority voting with higher-order aggregation. Our CGD uses zero-shot inter-model agreement — requiring no training data.

4. The ARRIVAL Protocol

4.1 Atom Taxonomy

The protocol employs 66 structured semantic @-atoms from DEUS.PROTOCOL v0.5:

Core Atoms (46)

Identity (`@SELF`, `@OTHER`), Goals (`@GOAL`), Reasoning (`@REASON`, `@EVIDENCE`, `@HYPOTHESIS`), Conflict (`@CONFLICT`, `@DISAGREE`), Consensus (`@CONSENSUS`, `@RESOLUTION`), Meta (`@META`, `@REFLECT`).

Promoted Emergent Atoms (20)

Discovered in Phase 4 experiments (5/5 architecture adoption, frequency ≥ 5): Planning (`@ACTION_PLAN`, `@PROPOSAL`), Evaluation (`@EVALUATION`, `@SYNTHESIS`), Risk/Knowledge (`@RISK_TOLERANCE`, `@UNCERTAINTY`).

4.2 Four-Round Protocol

1. **R1 — Independent Analysis:** Each agent independently analyzes the question using `@HYPOTHESIS`, `@EVIDENCE`, `@CONFIDENCE`.
2. **R2 — Adversarial Peer Review:** Each agent reviews others' R1 responses using `@CONFLICT`, `@COUNTER`, `@DISAGREE`.
3. **R3 — CRDT Overlay:** Zero-cost mathematical reconciliation: CARE Resolve computes weighted consensus.
4. **R4 — Synthesis:** Designated synthesizer produces final consensus answer.

4.3 Confidence-Gated Debate (CGD)

Phase 22 introduces an agreement-gated variant:

CGD Protocol

1. **Independent Solo** — All K models answer independently (no inter-model exposure).
2. **Agreement Check** — Classify: unanimous (K/K), split 2v1, split 3-way.
3. **Targeted Debate** — Only when disagreement exists; final answer by majority vote (no R4).

This eliminates anchoring bias and R4 regression problems.

5. MEANING-CRDT v1.1

We formalize conflict resolution through MEANING-CRDT v1.1 [16], a mathematical framework based on Conflict-free Replicated Data Types [27].

5.1 CARE Resolve

Theorem 1 (Optimality)

CARE Resolve minimizes total weighted semantic distance:

$$\hat{\mathbf{v}} = \arg\min_{\mathbf{v}} \sum_{i=1}^{N} w_i \, d(\mathbf{v}, \mathbf{v}_i)^2 = \frac{\sum_i w_i \mathbf{v}_i}{\sum_i w_i}$$

Theorem 3 (Factor-4 Improvement)

Under mild conditions, CARE provides up to 4× reduction in total semantic distance compared to unweighted averaging.

Theorem 5 (Convergence)

Iterated CARE converges to a fixed point in $O(N)$ rounds for N agents with bounded weight ratios.

5.2 Meaning Debt

Theorem 6 (Bounded Debt)

Under the ARRIVAL protocol, Meaning Debt is bounded: $\mathrm{MD} \leq w_{\max} \cdot D_{\max} \cdot N$.

Theorem 8 (Bayesian Equivalence)

CARE Resolve is equivalent to Bayesian belief updating when confidence weights correspond to prior precision parameters.

5.3 Vulnerability Analysis

Theorem 9 (Weight Inflation)

An adversarial agent can exploit CARE by inflating its confidence. The damage is bounded by $w_{\mathrm{adv}} / \sum w_i$, making CARE robust when honest agents' total weight dominates.

6. Experimental Setup

6.1 Models and Infrastructure

We evaluate across 17 LLM architectures (Table 1).

Table 1: Models used across 23 experimental phases.

Model	**Provider**	**Params**	**Phases**
GPT-4o	OpenAI	Undisclosed	4–12
Claude 3.5 Sonnet	Anthropic	Undisclosed	4–12
DeepSeek V3	DeepSeek	671B MoE	4–13
DeepSeek R1	DeepSeek	671B MoE	4–13
Llama 3.3 70B	Meta	70B	4–12
Qwen 2.5 72B	Alibaba	72B	4–12
Mistral Large	Mistral	Undisclosed	4–12
Gemini 2.0 Flash	Google	Undisclosed	4–12
Qwen3-235B	Alibaba	235B MoE	13, 16, 17
Claude Sonnet 4.5	Anthropic	Undisclosed	18
GPT-4.1	OpenAI	Undisclosed	18, 20–22
DeepSeek V3.2	DeepSeek	MoE	18, 23
Mistral Large 3	Mistral	Undisclosed	18
Gemini 3 Flash	Google	Undisclosed	18, 20–23
Grok 4.1 Fast	xAI	Undisclosed	18, 20–23
Claude Sonnet 4.6	Anthropic	Undisclosed	23
Kimi K2.5	Moonshot AI	Undisclosed	23
GLM-5	Zhipu AI	Undisclosed	23
Qwen3.5-397B	Alibaba	397B MoE	23

All models accessed via OpenRouter API. No model fine-tuning; all coordination via system prompt injection.

6.2 Statistical Methods

All pairwise accuracy comparisons use **McNemar's test** (exact, two-sided) for matched pairs on the same question set. **Fisher's exact test** is used when comparing independent conditions (e.g., GovSim survival rates). Confidence intervals are **Wilson score** intervals at 95%. Effect sizes are reported as accuracy differences in percentage points (pp). Where applicable, **Mann–Whitney U** is used for non-parametric comparisons of continuous metrics (CARE scores, Meaning Debt). No correction for multiple comparisons is applied; all p-values are reported for transparency.

6.3 Benchmark

GPQA Diamond [25]: graduate-level science, resistant to web search. Phases 13/16/17 use 40 questions; Phases 20–23 use the full 198-question set (physics 86, chemistry 93, biology 19).

Current GPQA Diamond SOTA: 94.1% (Gemini 3.1 Pro [11]). Human expert accuracy: 69.7%. As of March 2026, multiple frontier models exceed 90%: Gemini 3.1 Pro 94.1%, GPT-5.3 Codex ∼91.5%. Kim et al. [17] investigate scaling laws in multi-agent systems.

6.4 Budget

Table 2: Cost breakdown across all 23 experimental phases.

Phase	Experiments	API Calls	Cost (USD)
4–5	485	∼2,000	∼$2.00
6–12	∼80	∼500	∼$1.50
13	80	∼320	∼$1.00
14–15	∼40	∼200	∼$0.50
16	200+	640	$1.92
17	200	200	$0.50
18	6	∼66	$0.77
19	25	2,165	$5.76
20	198	∼1,782	$7.43
21	792	∼1,386	$9.63
22	198	∼692	$5.72
23	198	∼1,822	$11.35
AutoGen	16	∼100	$0.02
Total	**2,200+**	**∼14,500**	**<$95**

7. Results: Core Protocol (Phases 4–12)

In 385 dyadic coordination experiments across 8 LLM architectures, cross-architecture consensus reached **98.6%** (142/144 unique pairings). Models spontaneously generated 506 emergent atoms beyond the standard 46, with 1,173 cross-architecture adoptions.

Adversarial testing (Phases 6–7): Byzantine saboteurs with `@SABOTAGE` atoms were detected and neutralized by honest agents in >90% of cases. Trojan atoms (designed to pass validation) were identified and rejected within 2 rounds.

8. Results: GPQA Diamond (Phase 13, N=40)

Phase 13 Key Result

ARRIVAL 63.8% vs. Majority Voting 42.5% (McNemar $p = 0.006$, significant on 40 questions).

Per-domain: Biology +40 pp (80% vs. 40%), Chemistry +24 pp (57% vs. 33%), Physics +5 pp (57% vs. 52%).

9. Results: Homogeneous Ensemble (Phase 16)

Phase 16 tests a 5-copy Qwen3-235B ensemble with different personas. Result: **65.0%** (26/40) vs. baseline MV 52.5% (+12.5 pp, McNemar $p = 0.006$). Echo-chamber analysis reveals mean uniqueness ratio of 0.73 (sufficient diversity despite identical architecture).

10. Results: Solo CoT Baseline (Phase 17)

Solo Qwen3-235B with enhanced CoT: **70.0%** MV accuracy, 5 pp above ARRIVAL's 65.0% (not significant, $p = 0.812$). Validates Wang et al.'s finding that strong single-agent prompting can match multi-agent on accuracy.

11. Results: Full GPQA Diamond (Phases 20–23)

11.1 Phase 20: Mid-Tier Models (N=198)

Phase 20 Key Result

ARRIVAL MV **67.7%** vs. Solo CoT MV 62.1% (+5.6 pp, McNemar $p = 0.233$, ns).

Per-domain breakdown:

Table 3: Phase 20 per-domain accuracy (N=198).

Domain	Arrival MV	Solo CoT MV	Δ
Biology (19)	84.2%	63.2%	+21.1 pp
Chemistry (93)	64.5%	58.1%	+6.5 pp
Physics (86)	67.4%	66.3%	+1.2 pp
Overall (198)	**67.7%**	**62.1%**	**+5.6 pp**

11.2 Phase 21: Frontier Models (N=198)

Phase 21 Key Result — Debate Effect Inverts

Non-Debate MV **82.3%** vs. ARRIVAL MV 78.8% (−3.5 pp, $p = 0.265$, ns).
Anchoring: Gemini drops from 82.3% (solo) to 78.3% (debate) — a 4.0 pp degradation.

R4 finalizer is consistently net negative: Phase 20 (9 regressions vs. 7 rescues), Phase 21 (13 vs. 8).

Table 4: Phase 21 voting strategy comparison (N=198).

Strategy	Accuracy
Oracle (any of 3 correct)	93.9%
Non-Debate MV	82.3%
Grok-weighted MV	86.4%
Solo Grok 4.1 Fast	85.4%
Solo GPT-4.1	78.3%
Solo Gemini 3 Flash	76.3%
ARRIVAL R4	76.3%
ARRIVAL MV	78.8%

11.3 Phase 22: Confidence-Gated Debate (N=198)

Phase 22 Key Result — CGD

CGD achieves **86.4%** accuracy (171/198).
95% CI: [80.9%, 90.5%].
vs. ARRIVAL MV: +7.6 pp (McNemar $p = 0.009$, significant).
61.6% questions resolved unanimously (95.9% accuracy).
Cost: \$5.72 for 198 questions (\$0.029/q).

Table 5: Phase 22 CGD results on GPQA Diamond (198 questions).

Method	Correct	Accuracy	Notes
CGD (Phase 22)	171/198	**86.4%**	Primary result
Solo Grok 4.1 Fast	169/198	85.4%	Best solo
Non-Debate MV	163/198	82.3%	Phase 21
ARRIVAL MV	156/198	78.8%	Phase 21
Oracle (any of 3)	189/198	95.5%	Diversity ceiling

Table 6: Phase 22 CGD per-domain accuracy.

Domain	Correct/Total	Accuracy
Physics (86)	82/86	**95.3%**
Chemistry (93)	74/93	79.6%
Biology (19)	15/19	78.9%
Overall (198)	**171/198**	**86.4%**

Minority-Was-Right: In 65 split 2v1 questions, minority correct 15 times (23.1%). GPT-4.1 as minority: 1/32 = 3%. Grok: 8/18 = 44%.

11.4 Phase 23: 7-Model Cross-Vendor CGD

Table 7: Phase 23 model lineup: 3 US + 4 Chinese models from 5 vendors.

#	Model	Vendor	Country	Solo
1	Grok 4.1 Fast	xAI	US	**87.4%**
2	Gemini 3 Flash	Google	US	82.8%
3	Claude Sonnet 4.6	Anthropic	US	80.8%
4	DeepSeek V3.2	DeepSeek	CN	74.7%
5	Kimi K2.5	Moonshot AI	CN	72.2%
6	GLM-5	Zhipu AI	CN	70.7%
7	Qwen3.5-397B	Alibaba	CN	58.6%

Phase 23 Key Result — CGD-7

CGD-7: **86.9%** (172/198) vs. Phase 22 CGD: 86.4% (+0.5 pp, $p = 1.0$, ns).
Best solo (Grok): 87.4% — solo > ensemble.
Oracle ceiling: 96.5% (191/198).
Cost: $11.35 total.

11.4.1 Debate Type Analysis

Table 8: CGD-7 debate type breakdown (198 questions).

Type	N (%)	Accuracy	Action
Unanimous (7/7)	105 (53%)	99.0%	Locked
Supermajority (6/7)	23 (12%)	82.6%	Locked
Strong maj. (5/7)	22 (11%)	81.8%	Locked
Simple maj. (4v3)	21 (11%)	66.7%	Minority debate
No majority	27 (14%)	63.0%	Full debate

75.8% of questions locked without debate (≥5/7 agreement), achieving 93.3% accuracy.

11.4.2 Ablation: Weighting, Pruning, Poisoned Voters

Weighted vs. Simple: Weighting *hurts* — Weighted MV (85.4%) < Simple MV (85.9%) < CGD-7 (86.9%).

WMV+D-3 (pruned 4-model subset): Dropping the 3 weakest Chinese models yields **88.4%** (175/198) — the project-high accuracy.

Poisoned voter: Qwen3.5-397B exhibits 5% minority-correct rate with 54 extraction failures (27%). Its dissent reliably indicates the *wrong* answer.

Thinking mode confound: US models averaged 83.7% vs. CN 69.1% — but reasoning was disabled for all Chinese models, making this comparison confounded.

12. Results: Applied Tasks (Phase 18)

Phase 18: Five Models Match One Frontier Model

5-model ARRIVAL swarm matches Claude Sonnet 4.5:
Security audit: 10/10 bugs found (both).
Code generation: 53% vs. 60% test pass rate.
Cost: $0.14 vs. $0.12 per task.

CARE as task discriminator: analytical tasks (security audit) produce CARE ≈ 0.50; constructive tasks (code generation) produce CARE ≈ 0.97.

13. Results: Memory and CARE-ALERT (Phases 14–15)

Hypercorrection paradox (Phase 14): Naive memory injection degrades accuracy by 5.7 pp.

Gated CARE-ALERT (Phase 15): Monitors Meaning Debt in real time; restores baseline performance (CARE improvement $p = 0.042$). Validated in Phase 18: correctly activates when CARE = 0.50 and suppresses when CARE = 0.97.

14. Results: AutoGen Validation

Framework-agnostic validation on Microsoft AutoGen (AG2) [33] achieves 100% behavioral match with the reference implementation across 16 test cases.

15. Results: GovSim (Phase 19)

Phase 19: Cooperation Breakthrough

ARRIVAL converts **0% baseline survival to 100% survival** across all 25 runs (5 conditions $\times$ 5 runs).
Fisher exact $p = 0.008$.
CARE = 0.97, Gini = 0.001 (near-perfect equality).
3.2$\times$ resource extraction vs. baseline collapse.

Key findings:

1. **Self-healing**: Greedy proposals neutralized within 1 round via CARE convergence.
2. **Condition D (binding-only)**: 100% survival at 32$\times$ lower cost.
3. **Condition E (adversarial R4)**: Survival preserved (safety clamp) but fairness destroyed (Gini = 0.800).
4. **Empirical support for Theorems 1, 5, 6, 8**.

16. Discussion

16.1 What ARRIVAL Adds Beyond Strong Prompting

ARRIVAL does not claim to always improve raw accuracy. Its value lies in: (1) structured reasoning traces, (2) error detection via peer review, (3) formal guarantees via MEANING-CRDT, (4) scalability, and (5) protocol evolution — CGD emerged from Phase 20–21 failure analysis, achieving 86.4%.

16.2 Cost-Efficiency

At under \$95 for ∼14,500 API calls across 23 phases and 17 architectures, ARRIVAL demonstrates exceptional cost-efficiency. Phase 22 CGD: \$5.72 for 198 questions (\$0.029/q). Phase 23 CGD-7: \$11.35 for 198 questions (\$0.057/q). Multi-agent overhead: ∼\$0.012–\$0.057 per question.

16.3 CGD: From Failure Analysis to Protocol Innovation

CGD addresses three Phase 20–21 problems: (1) anchoring from sequential exposure, (2) destructive R4 overrides, (3) false consensus. Grounded in the Condorcet Jury Theorem: independent voters with accuracy $> 50\%$ produce better collective decisions. Choi et al. [6] prove that debate has a martingale property — CGD sidesteps this by reserving debate for genuine disagreement.

16.4 Quality Threshold and Non-Monotonic Scaling

Phase 23 reveals non-monotonic scaling: 3→7 models yields only +0.5 pp ($p = 1.0$), while pruned 4-model WMV+D-3 achieves 88.4%. Qwen3.5-397B: 5% minority-correct, 27% extraction failures — a **poisoned voter**. Implication: model selection > model count.

16.5 Thinking Mode as Hidden Variable

US–CN accuracy gap (83.7% vs. 69.1%) is confounded by systematic disabling of reasoning modes for Chinese models. The 14.6 pp gap should **not** be interpreted as inherent architectural differences.

17. Limitations

1. **Benchmark specificity**: GPQA Diamond results may not generalize to all reasoning tasks.
2. **API dependence**: All experiments use API-accessible models; local/fine-tuned models untested.
3. **Sequential bias**: The 4-round structure can introduce anchoring (addressed by CGD).
4. **Cost of debate**: Multi-agent overhead may not be justified when a single frontier model suffices.
5. **CARE sensitivity**: Heuristic weight extraction from confidence declarations.
6. **No social agency**: LLM agents lack genuine social agency [19].
7. **Homogeneous caveat**: Phase 16 uses 5 copies of the same model.
8. **Phase 18 N=1**: Applied experiments lack repeated trials.
9. **Code synthesis**: 4-round protocol incurs coherence loss.
10. **Single game type**: GovSim fish pond only.
11. **No adversarial agents**: Cooperative personas only.
12. **CGD limited scope**: Tested on MCQ with $K = 3$ and $K = 7$; non-MCQ untested.

18. Conclusion and Future Work

We presented the Arrival Protocol, demonstrating that structured semantic communication enables multi-agent LLM coordination across benchmarks, practical tasks, and cooperation scenarios. Key results:

- **Phase 22 CGD**: 86.4% on GPQA Diamond (171/198), +7.6 pp vs. Arrival MV ($p = 0.009$).
- **Phase 23 CGD-7**: 86.9% with 7 models; non-monotonic scaling confirmed.
- **WMV+D-3**: 88.4% (175/198) — project-high accuracy.
- **Phase 19**: 0%→100% survival in GovSim ($p = 0.008$).
- **Phase 18**: 5 cheaper models match frontier solo.

Honest picture: standard Arrival does not significantly outperform solo models on MCQ at scale; with frontier models, debate hurts via anchoring. CGD demonstrates that failure analysis was productive: 86.4% is the highest in the project. Arrival's primary value: cooperation (GovSim), transparency, vendor independence, and principled ensemble aggregation.

Future Work

1. Extend CGD to additional benchmarks (MMLU-Pro, ARC-Challenge, MedQA).
2. CGD extensions: $K > 3$ models, weighted confidence gating.
3. Broader cooperation games: public goods, prisoner's dilemma.
4. Adversarial agents: RL-trained defectors.
5. Adaptive orchestration via real-time CARE measurements.
6. Scaling laws: accuracy vs. number of agents.
7. Online memory: cross-session coordination patterns.
8. Larger N for MCQ ($N \approx 600$) for statistical power.
9. CGD vs. frontier SOTA: cost-accuracy tradeoffs.

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A. DEUS.PROTOCOL v0.5 Atom Dictionary

See `docs/ATOM_DICTIONARY.md` in the repository for the complete specification of all 66 atoms with usage examples and adoption statistics.

B. MEANING-CRDT v1.1 Theorem Proofs

Full proofs for all theorems are available at DOI: 10.5281/zenodo.18702383

C. Experiment Logs and Reproducibility

All experiment configurations, API logs, and result files are available in the `experiments/` and `results/` directories. Total cost: under $95 USD across 2,200+ experiments and ∼14,500 API calls. See `docs/setup/SETUP_EN.md` for reproduction instructions.